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1. Types of Computers

A computer may be defined as: "*A device or set of devices that can store data and a program that operates on the data.*"

The first fully operational general-purpose computer was the **Z3**, built by **Konrad Zuse** in Germany in 1941 — electromechanical, using binary digits.

Type	Description	Status
Analogue	Uses continuous physical variables (voltage, pressure) to represent measurements. Used as electronic models of mechanical systems where experiments on the real system would be costly or dangerous (e.g., bridge/aircraft wing design).	Less common now — digital computers can simulate their function
Digital	Uses digital (binary) data — only two voltage levels: ON or OFF. Binary (base-2) number system. By far the most common type today.	Dominant — used in virtually all modern avionics

2. Digital Computer Components (Hardware)

```
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  A["Input Devices
(keyboard, mouse, sensors)"] --> B["CPU
Central Processing Unit"]
  B --> C["Output Devices
(monitor, printer, display)"]
  D["Memory
(ROM / RAM)"] <--> B
```

Basic computer components:

1. Input peripherals (devices)
2. Central Processing Unit (CPU)
3. Memory
4. Output peripherals (devices)

Note: "Inertial unit" and "auto brightness control" are NOT computer components.

3. CPU — Internal Structure

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The CPU = the "brain" of the computer. It performs, organises and controls all operations. Three components:

1. **Arithmetic Logic Unit (ALU)** — performs arithmetic calculations and logical operations in the *binary number system*
2. **Shift Registers** — temporary stores; one of them, called the **accumulator**, contains the data actually being processed
3. **Control Unit** — contains the computer's clock (crystal-controlled oscillator at typically **120–45 MHz**); synchronises all computer operations

CPU does NOT include: input devices, output devices, hard disk, or BIOS. These are external to the CPU itself.

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4. Input and Output Devices

Type	Examples (PC)	Examples (Aircraft)
Input peripherals	Keyboard, mouse, modem	Rad Alt, Baro Alt, TAS sensor, fuel flow sensor
Output peripherals	Monitor (VDU), printer	EFIS Symbol Generator, FMS CDU display

BIOS (Basic Input Output System) — converts input signals to a form the computer can work with, and converts outputs to a form the operator or another aircraft system can understand.

Classification rules:

- Keyboard = INPUT
- Mouse = INPUT
- Modem = INPUT (and output, but classified as input peripheral here)
- Screen/monitor = OUTPUT
- Printer = OUTPUT
- Hard disk / floppy disk = STORAGE (both read and write — classified as memory, not strictly input or output)

5. Memory — Volatile and Non-Volatile

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Memory Type	Description	Volatile?
ROM — Read Only Memory	Programmed by manufacturer; cannot be altered by user. Used for fixed programs.	Non-volatile
PROM — Programmable ROM	User can program it once	Non-volatile
EPROM — Erasable PROM	Can be erased (UV light) and reprogrammed	Non-volatile
EEPROM — Electrically Erasable PROM	Can be erased electronically and reprogrammed	Non-volatile
RAM — Random Access Memory	Read and write at will; used for data currently being processed; contents lost when power removed	**Volatile**

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Volatile vs. Non-volatile:

- **Volatile memory** = loses data when power is removed (e.g., RAM)
- **Non-volatile memory** = retains data when power is removed (e.g., ROM, EPROM, Flash memory)

RAM is normally volatile — if power fails, all data in RAM is lost.

Permanent Storage (Long-Term)

Type	Details	Re-writable?
Flash Memory (electronic chip)	Solid-state; very fast access	Yes — indefinitely
Optical Disk (CD)	Read with laser; read-only or re-writable	Limited rewrites only
Magnetic Disk (HDD)	Internal, external, or removable; "floppy disks" now obsolete	Yes — indefinitely

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6. Software — Programming Languages

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Level	Type	Description
Low	**Assembly Language**	Symbolic representation of machine codes. Defined by the hardware manufacturer. Close to machine language (binary/hex).
High	**High Level Language**	Not limited to a specific computer or job. Easier to understand. Examples: **FORTRAN, ALGOL, BASIC, C++**.
—	**Scripting Language**	Controls one or more applications. Written in a different language (similar to BASIC). Created or modified by end-user. Applies to a specific program only.

7. EUROCAE ED12B — Aircraft Software Safety Levels

All aircraft-related software must comply with **EUROCAE ED12B** (also known as DO-178B/C). Five software levels:

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Level	Failure Condition	Aviation Context	Example
A	Catastrophic	Prevents continued safe flight or landing	Flight Control Computer
B	Hazardous / Severe-Major	Potential fatal injuries to a small number of occupants	Flight Management Computer
C	Major	Impairs crew efficiency; discomfort or possible injury to occupants	Navigation displays
D	Minor	Reduced safety margins but well within crew capabilities	Cabin systems
E	No Effect	Does not affect safety of the aircraft at all	Galley oven controller

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Key principle: The required software rigour level depends on what happens if that software fails. The galley oven can have a major failure without endangering the aircraft (Level E), whereas a Flight Control Computer failure could be catastrophic (Level A).

8. Aircraft Computer Systems

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Current computer-controlled aircraft systems:

- Flight Management System (FMS)
- Digital Flight Guidance System (DFGS)
- Ground Proximity Warning System (GPWS)
- Traffic Alert Collision Avoidance System (TCAS)
- Fly-by-wire flight control (entire flight envelope)

Current design: Dedicated computers for each separate system.

Future trend: Integrated Hazard Warning System (IHWS) — a powerful central processor handling inputs from stall warning, windshear detection, GPWS, TCAS, and weather radar; processes and prioritises warnings to crew.

9. Analogue-to-Digital and Digital-to-Analogue Conversion

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  A["Aircraft sensor
(analogue: voltage, pressure, temp)"] -->|"A/D Converter"| B["Digital Computer
(binary data)"]
  B -->|"D/A Converter"| C["Analogue display
or analogue device"]
```

- **A/D Converter (Analogue to Digital):** Many aircraft sensors produce analogue information (varying voltages, pressures, temperatures). The A/D converter is required in the interface between the sensor and the digital computer input device.
- **D/A Converter (Digital to Analogue):** When a digital computer must pass information to an analogue device (e.g., older analogue instruments), the process is reversed.

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Practice Questions & Detailed Answers

Ghost Aviator Instrumentation — Chapter 23. DGCA CPL/ATPL.

Q1. A basic digital computer consists of: (1) input peripherals (2) CPU (3) inertial unit (4) memory (5) auto brightness control (6) output peripherals

a. 1, 2, 3, 4 and 6

b. 1, 2, 4 and 6

c. 1, 4, 6 only

d. 2, 3, 4 and 6

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✅ Correct Answer: **B**

The four basic components of a digital computer are: (1) input peripherals, (2) CPU, (4) memory, and (6) output peripherals. Items (3) inertial unit and (5) auto brightness control are NOT computer components — they are aircraft-specific or display-specific hardware unrelated to the computer architecture.

Q2. The Central Processing Unit (CPU) consists of: (1) input device (2) output device (3) ALU (4) shift registers (5) control unit (6) hard disk

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a. 1, 2, 3 and 5

b. 3, 4, and 6

c. 1, 2, 5, and 6

d. 3, 4 and 5

✅ Correct Answer: **D**

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The CPU consists of: (3) ALU, (4) shift registers (including accumulator), and (5) control unit (with clock). Input devices, output devices, and hard disk are EXTERNAL to the CPU. The CPU is just the processing core.

Q3. In computer terminology an input peripheral device would be:

a. A hard disk

b. A floppy disk

c. A keyboard

d. A screen display unit

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✅ Correct Answer: **C**

A keyboard is a classic input peripheral — it sends data INTO the computer. Hard disk and floppy disk are storage devices (not strictly input or output). A screen display unit is an OUTPUT peripheral.

Q4. In computer terminology an output peripheral device would be:

a. A floppy disk

b. A hard disk

c. A screen display unit

d. A keyboard

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✅ Correct Answer: **C**

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A screen display unit (monitor/VDU) is an output peripheral — the computer sends processed data OUT to the screen for the user to see. Keyboard = input. Disks = storage. In an aircraft, the EFIS symbol generator output and the FMS CDU display are output devices.

Q5. In computer terminology a memory which loses its data when power is removed is called:

a. Non-volatile

b. Non-permanent

c. Non-retentive

d. Volatile

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✅ Correct Answer: **D**

Volatile memory loses its data when power is removed. RAM is the primary example of volatile memory. The word "volatile" means easily evaporated/lost — exactly what happens to RAM data when power goes. "Non-volatile" = retains data without power.

Q6. In computer terminology a memory which retains its data when power is removed is called:

a. Non-volatile

b. Volatile

c. RAM

d. ROM

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✅ Correct Answer: **A**

Non-volatile memory retains data when power is removed. Examples: ROM, EPROM, EEPROM, Flash memory, hard disks, optical disks. Note that ROM is one specific type of non-volatile memory, but the

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general term is "non-volatile." Choosing ROM (D) would be too specific when the question asks for the general terminology.

Q7. Examples of input peripheral devices are: (1) mouse (2) modem (3) printer (4) screen display unit (5) keyboard

a. 2, 3, 4 and 5

b. 1, 2 and 5

c. 1 and 5

d. 1, 2, 3

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✅ Correct Answer: **B**

Input peripherals: (1) mouse, (2) modem, (5) keyboard — these all send data INTO the computer. Printer (3) and screen display unit (4) are OUTPUT devices — they receive data FROM the computer to display or print.

Q8. In computer terminology "software" refers to:

a. The memory system — floppy disks, hard disks, etc.

b. The RAM and ROM capacity

c. The programme of instructions

d. The BIOS

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✅ Correct Answer: **C**

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Software = the programme of instructions. This is the fundamental distinction: Hardware = physical components; Software = the programs/code that run on the hardware. BIOS is itself software (a specific type of software), but the general definition of software is the programme of instructions.

Q9. In computer terminology "hardware" refers to:

a. The digital computer components — keyboard, monitor, CPU, etc.

b. The permanent memory system and its capacity

c. The RAM capacity

d. The programme of instructions

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✅ Correct Answer: **A**

Hardware = all the physical components of the computer system: keyboard, monitor, CPU, memory chips, disk drives, etc. The chapter title is "Digital Computer Components (Hardware)" — confirming that hardware = the physical components.

Q10. The permanent memory of a digital computer usually takes the form of:

a. Integrated circuits rated in megabytes

b. Shift registers whose capacity is rated in mega or gigabytes

c. Floppy or hard disks whose capacity is measured in mega or gigabytes

d. Central Processing Unit

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✅ Correct Answer: **C**

Permanent (long-term) storage takes the form of magnetic disks (hard disks, floppy disks — though floppies are now obsolete), optical disks, or flash memory chips — all measured in mega or gigabytes. Shift registers are temporary stores within the CPU (volatile). The CPU itself is not a storage medium.

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Note: flash memory is an electronic chip but the book specifically mentions floppy/hard disks for "permanent storage."

Q11. The purpose of the Arithmetic Logic Unit within the CPU is to:

a. Act as a temporary store for information being processed

b. Perform calculations in the binary number system

c. Perform calculations in the binary, octal or hexadecimal system

d. Perform all clock functions based on the computer clock frequency

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✅ Correct Answer: **B**

The ALU performs arithmetic calculations and logical operations in the binary number system. The shift registers (including accumulator) act as temporary stores — not the ALU. The Control Unit handles the clock functions.

Why the others are wrong:

- **A:** Temporary storage = function of shift registers / accumulator.
- **C:** The book says "binary number system" specifically — not binary, octal and hexadecimal.
- **D:** Clock functions = Control Unit (which contains the crystal oscillator clock).

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Q12. Within the CPU, the temporary stores and accumulator which handle data during processing are called:

a. Arithmetic Logic Unit (ALU)

b. Shift registers

c. Control unit

d. BIOS

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✅ Correct Answer: **B**

Shift registers are the temporary stores within the CPU. One of the shift registers, specifically called the accumulator, contains the data actually being processed at any given moment. ALU does the calculations; Control Unit provides timing; BIOS is the input/output conversion system external to the CPU.

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